Studying LLM-Based Political Forecasting under Biased Media Intake

Cade Harger
[cade.harger@gmail.com]

Abstract

[ABSTRACT PLACEHOLDER]

1 Introduction

Recently, LLMs (Large Language Models) and their ability to consistently process unstructured textual data from almost any source have led them to be used in forecasting systems where relevant predictor data isn't centralized in a tabular format. This contrasts with typical predictive models that train on clearly defined and normalized rows and columns of data.

One domain forecasting LLMs have been used for (LRL) is politics. Political event prediction is a broad domain, where events might include elections, wars, trade deals, and many more sub-domains. LLMs are especially well-suited to this domain for a number of reasons. While structured data exists to predict such events, the highly nuanced nature of politics means that often relevant data isn't easily captured into a small number of variables that can be loaded into a database for a downstream predictive model. This makes unstructured data like natural language text, if properly utilized, more useful. Additionally, some political events are regularly repeated with clearly defined historical data, but many others are somewhat unprecedented, such as the COVID-19 pandemic and resulting political decisions. This results in an environment where forecasting techniques that can generalize well to new scenarios will do well.

Some of these LLM forecasting systems use search engine results as their primary source of input data, which turns out to be effective [**?**]. For political events, search engine queries are likely to return news articles. Search engines, by nature, pick up results from more popular outlets. For US politics, many of the top political news outlets [**?**] are known for having some partisan bias [**?**]. This raises questions of how this bias might affect LLM forecasting systems. If this effect is shown to result in bias in the output of such systems, then it would call for future research into mitigation strategies in order to preserve the efficacy of these systems when implemented in decision-making networks. It would also pave the way for bias-measurement systems, where the change in a forecasting system's output given a piece of media could be used as a semi-objective measure of bias.

While the effects of consuming biased media directly upon the human-led forecasting of political event outcomes has not been extensively studied, the effects of consuming biased media has been shown to be correlated with biased perceptions of factors such as the economy [**?**] (reword for causality?). The underlying truth of these factors is important information for making effective political decisions, and they are predictors in automated tabular political forecasting systems; therefore, they should logically propagate into the human forecasting decision-making process, leading to biased predictions. The only alternative interpretation is that the belief distortions documented in **?**] and others do not propagate into predictive reasoning. This itself would defy conventional knowledge of human decision making [**?**] and be a significant finding about the relationship between media-induced beliefs and political forecasting.

In this work, we study a simple LLM-based U.S. political event forecasting system that sources input data for predictions through a search engine. We examine the associations between the inclusion of news articles from outlets that are commonly accepted to be biased in a partisan direction and the system's output predictions. We show that, under this experiment's given circumstances, bias does not appear to propagate to the LLM system's prediction efficacy in a statistically significant way. This contrasts with the human behavior in the previous assumption that bias from political media consumption propagates to political outcome prediction. This finding invites future research into what causes this association and why it is different from the human assumption.

2 Previous Work

[PREVIOUS WORK PLACEHOLDER]

3 Method

3.1 Overview

We construct an automated pipeline to measure the association between partisan news media and the output predictions of an LLM-based U.S. political event forecasting system. The pipeline collects settled political prediction market events, gathers news articles via a search engine, applies leak detection to exclude articles that reveal the outcome, and then queries an LLM to predict each event's outcome under controlled article conditions. We compare predictions made with articles from outlets with known partisan leanings against a control condition using only non-bias-site articles, and analyze the resulting prediction shifts using various statistical methods.

3.2 Event Data Source: Political Prediction Markets

We use Kalshi, a regulated prediction market exchange, as our source of ground-truth political event outcomes. We collect settled binary markets from Kalshi's Politics category. We filter to events with exactly one market (a single yes/no question) that have been resolved to either "yes" or "no." At most one event per Kalshi series is retained to avoid redundant events from the same topic. The result dataset has 399 events whose resolution date falls on or after October 1, 2024. This date was chosen to be well after the cutoff date of June 1st, 2024, of the model used for prediction, OpenAI's GPT-4.1, in order to prevent data leakage.

For each event, the market title serves as the prediction question (e.g., "Will the Republican candidate win the Senate race in X?"), and the resolved outcome (yes/no) provides the ground-truth label.

Each event outcome is labeled with a partisan lean using GPT-5-mini with structured JSON output. Given the market title, the model classifies whether the "Yes" outcome would more likely benefit Republicans, Democrats, or is politically neutral. The classification prompt instructs the model to consider what "Yes" resolving means for each party's political interests and to return one of three labels: "republican," "democrat," or "neutral." This labeling is used in downstream analyses to examine whether bias-site effects differ by partisan direction. We treat this label as a zero-sum game pitting republicans against democrats, where if the "Yes" outcome benefits one party, the "No" outcome is set to benefit the other party. The LLM is made aware of this distinction. In the resulting dataset, the "Yes" outcome is designated to represent the Republican-favoring outcome 172 times, the Democrat-favoring outcome 92 times, and the neutral outcome 136 times. While there is a clear imbalance in these classes, the zero-sum game framing ensures republican and democrat classes are equally represented. The only confound would be if the LLM was biased towards either a "Yes" or "No" outcome, but we find no evidence for this.

3.3 Article Collection

For each event, we generate a targeted search query using GPT-5-mini and fetch up to 25 news articles from Exa, a neural search engine. Articles are restricted to those published at least 30 days before the event's resolution date (the "cutoff date"), ensuring that the article set represents information available well before the outcome was known. Exa returns full article text along with metadata, including URL, title, published date, domain, and author.

3.4 Leak Detection

Because search engine date filters are imperfect, some articles may reference or strongly imply the event outcome despite being nominally published before the cutoff. We apply LLM-based leak detection using GPT-5-mini in a dedicated filtering pass. First, for each event, a leak schema describing what sorts of information might indicate a leak is generated from the market title and cutoff date. Then, each article is evaluated against the event question, schema, and cutoff date; articles judged to reveal the outcome are removed from the dataset. This step removes roughly 2–3 articles per event.

Initially, we did not include any schema in the article leak detection. After manually inspecting 10 events to see if leak detection was effective, we found that several leaks passed the LLM's inspection. After adding the schema generation, we found zero leaks surviving filtering with no increase in false positives compared to evaluation with no schema for the manually inspected events, marking a significant improvement and demonstrating effective leak filtering.

3.5 Bias Site Selection

After the initial article collection, we analyze domain frequency counts across all collected articles. From this analysis, we select four news outlets as bias test sites based on their high event coverage (appearing across many events) and their commonly recognized partisan positioning: The New York Times (nytimes.com), CNN (cnn.com), New York Post (nypost.com), and Fox News (foxnews.com). The New York Times and CNN are generally characterized as left-leaning outlets, while the New York Post and Fox News are generally characterized as right-leaning outlets.[1]

3.6 Targeted Bias Article Collection

The initial broad search yields articles from the selected bias sites for only a small subset of events. To ensure higher coverage, we perform a targeted secondary search for each event using Exa's `includeDomains` filter, restricting results to the four bias sites. For each event, a single search returns up to 15 results, from which the first article per bias site that passes LLM leak detection is retained and appended to the event's article collection. This step increases per-site event coverage to 78.2% on average across the four sites.

3.7 LLM Prediction

We use GPT-4.1 as the prediction model. For each event, the model is prompted with the event question and a set of news articles that depend on the experiment stage, and asked to output a structured JSON response containing a brief explanation and a probability estimate $P(\text{YES}) \in [0, 1]$. All predictions use `temperature=0` to reduce noise.

We run two experimental configurations:

Control condition: The model receives the first 8 non-bias-site articles (excluding articles in the set of four bias sites) from the Exa filtered search results as context for its prediction. This serves as the baseline prediction reflecting a mix of articles that is relatively neutral and favors popular media sites.

Bias-site conditions: For each of the four bias sites, the model receives only the single article from that site and no other articles as context for its prediction. This isolates the individual influence of each bias-site article on the prediction.

3.8 Statistical Analysis

We compute the following metrics for each event-condition pair:

Delta: The change in predicted probability relative to the control condition, defined as $\delta = P_{\text{site}} - P_{\text{control}}$ for each bias site. A positive δ means the bias-site article pushed the prediction toward "Yes" relative to the control.

[1] `https://www.allsides.com/media-bias/media-bias-chart`

Brier Score: The squared error between the predicted probability and the actual binary outcome, $(P - \text{actual})^2$. Lower Brier scores indicate more accurate predictions.

Our analysis proceeds across several dimensions, each corresponding to a figure in the results:

1. **Relative bias vs. control (Figure 1, Panel 1):** Mean δ and 95% confidence intervals per bias site, showing the average prediction shift induced by each site's articles.
2. **Prediction accuracy (Figure 1, Panel 2):** Mean Brier score per condition with 95% confidence intervals, showing whether any site improves or degrades prediction quality relative to the control.
3. **Neutral event influence (Figure 2):** Mean δ for neutral-classified events only, isolating whether a site shifts predictions on events with no inherent partisan direction.
4. **Difference-in-Differences (Figure 3):** For each bias site, we compute $\text{DiD} = \text{mean}(\delta)_{\text{democrat}} - \text{mean}(\delta)_{\text{republican}}$. A positive DiD indicates that the site's effect on democrat-leaning events is more positive (or less negative) than its effect on republican-leaning events—i.e., the site shifts predictions toward democrat-favored outcomes relative to republican-favored outcomes. We report 95% confidence intervals and one-sided z-tests for $\text{DiD} > 0$ and $\text{DiD} < 0$.
5. **Base-rate-normalized analysis (Figure 4):** To address the potential confound that republican-leaning and democrat-leaning events may have different baseline probabilities (mean control $P(\text{YES})$), we compute a normalized delta: $\delta_{\text{norm}} = \delta \times (2 \times \text{lean\_binary} - 1)$, where $\text{lean\_binary} = 1$ for republican events and 0 for democrat events. This flips the sign convention so that positive δ_{norm} always means the site pushed predictions toward the "correct partisan direction."

All confidence intervals are computed as mean $\pm 1.96 \times$ SEM using the sample standard deviation with Bessel's correction (`ddof=1`). One-sided p-values for DiD tests are computed using the standard normal CDF.

4 Results

4.1 Relative Bias vs. Control

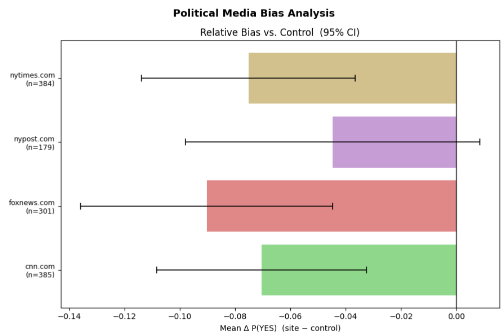
Across all four sites, the mean deltas are negative, implying that the bias site predictions relative to the controlpush predictions toward the "No" outcome on average. nytimes.com shows a mean δ of -0.0751 (95% CI: $[-0.1138, -0.0365]$, $n = 384$), cnn.com shows -0.0703 (95% CI: $[-0.1082, -0.0324]$, $n = 385$), nypost.com shows -0.0448 (95% CI: $[-0.0980, +0.0084]$, $n = 179$), and foxnews.com shows -0.0903 (95% CI: $[-0.1358, -0.0447]$, $n = 301$). The confidence intervals for nytimes.com, cnn.com, and foxnews.com entirely exclude zero, indicating a statistically significant downward shift in $P(\text{YES})$ for those three sites. The nypost.com interval spans zero, indicating no significant directional effect for that site.

4.2 Prediction Accuracy

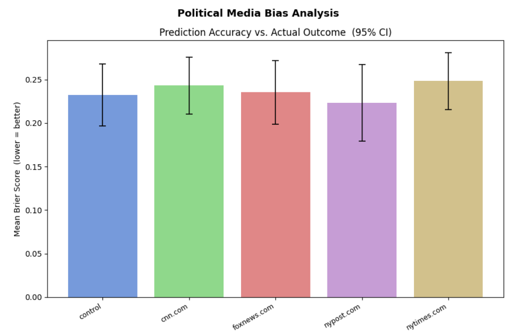
The control condition achieves a mean Brier score of 0.2324. The bias-site conditions achieve mean Brier scores of 0.2483 (nytimes.com), 0.2433 (cnn.com), 0.2231 (nypost.com), and 0.2353 (foxnews.com). The differences are small, with overlapping 95% confidence intervals across all conditions.

4.3 Neutral Event Influence

The mean deltas on neutral events are -0.1331 (95% CI $[-0.2085, -0.0578]$, nytimes.com, $n = 127$), -0.1019 (95% CI $[-0.1689, -0.0350]$, cnn.com, $n = 132$), -0.0461 (95% CI $[-0.1412, +0.0490]$, nypost.com, $n = 65$), and -0.1302 (95% CI $[-0.2140, -0.0464]$, foxnews.com, $n = 104$). Three of the four sites show confidence intervals entirely below zero,, indicating a systematic downward shift in $P(\text{YES})$ even on politically neutral events. This suggests a general directional bias toward "No" outcomes rather than a specifically partisan effect, as the shift persists in the absence of any partisan framing and visually matches the proportions of Figure 1 Panel 1.



(a) Relative bias vs. control.



(b) Prediction accuracy.

Figure 1: Aggregate prediction results. (a) Mean δ with 95% confidence intervals per bias site. (b) Mean Brier score with 95% confidence intervals per condition.

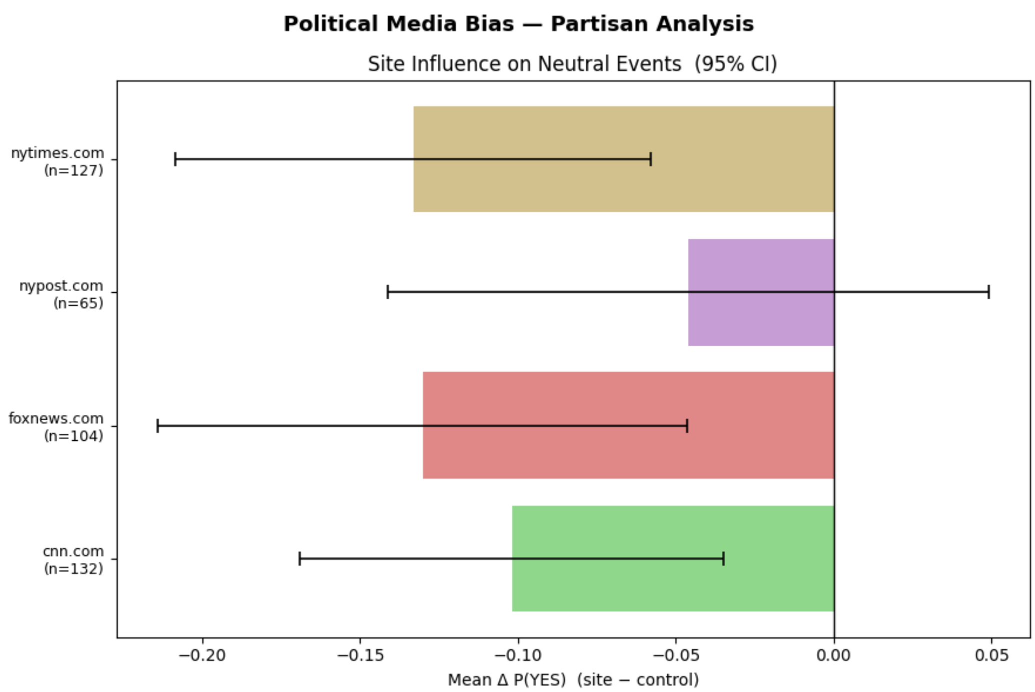


Figure 2: Mean δ on neutral-classified events per site, with 95% confidence intervals.

4.4 Difference-in-Differences

The DiD values, with 95% confidence intervals and one-sided p-values, are:

- nytimes.com: DiD $= +0.0496$, 95% CI $[-0.0378, +0.1369]$, $p(>0) = 0.133$, $p(<0) = 0.867$ ($n_{\text{dem}} = 87$, $n_{\text{rep}} = 170$)
- cnn.com: DiD $= +0.0676$, 95% CI $[-0.0264, +0.1615]$, $p(>0) = 0.079$, $p(<0) = 0.921$ ($n_{\text{dem}} = 86$, $n_{\text{rep}} = 167$)
- nypost.com: DiD $= +0.0597$, 95% CI $[-0.0666, +0.1861]$, $p(>0) = 0.177$, $p(<0) = 0.823$ ($n_{\text{dem}} = 42$, $n_{\text{rep}} = 72$)
- foxnews.com: DiD $= +0.0588$, 95% CI $[-0.0598, +0.1775]$, $p(>0) = 0.166$, $p(<0) = 0.835$ ($n_{\text{dem}} = 67$, $n_{\text{rep}} = 130$)

No site achieves statistical significance at the $p < 0.05$ level for DiD > 0, and all 95% confidence intervals include zero.

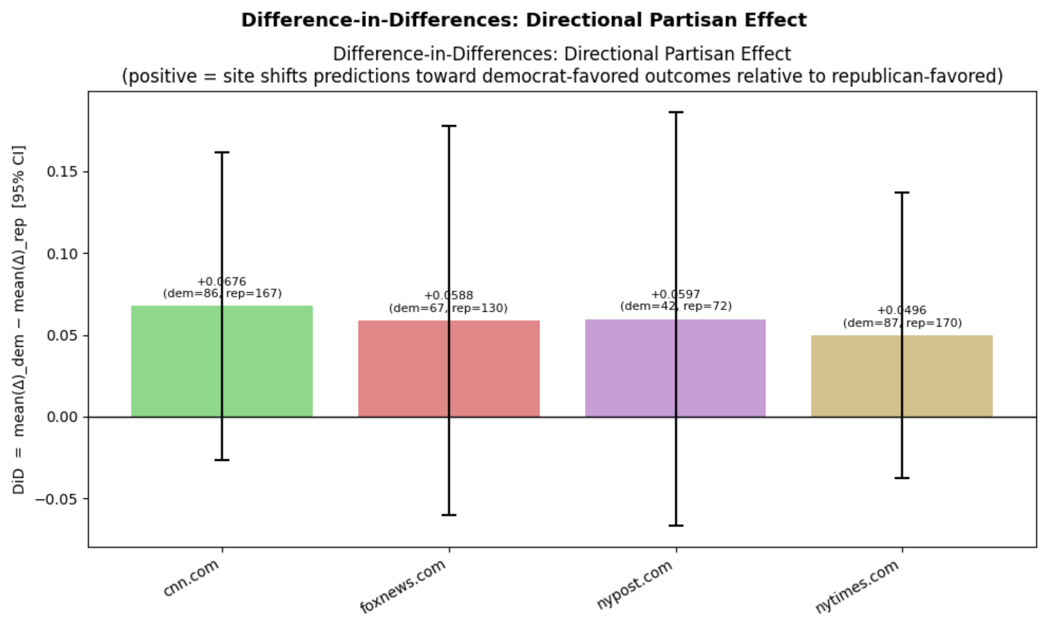


Figure 3: Difference-in-Differences (DiD) estimates for each bias site, with 95% confidence intervals. A positive DiD indicates the site shifts predictions toward democrat-favored outcomes relative to republican-favored outcomes.

4.5 Base-Rate Confound Check

Before interpreting the DiD results, we verify that republican-leaning and democrat-leaning events have comparable baseline prediction levels. The mean control $P(\text{YES})$ is 0.4208 ($n = 172$) for republican-leaning events and 0.3353 ($n = 91$) for democrat-leaning events, a difference of $+0.0855$. Since $|\Delta| = 0.0855 \geq 0.05$, there is a meaningful base-rate asymmetry between the two groups. This difference also is large enough to account for the observed DiD values. Republican-leaning and democrat-leaning events start from different $P(\text{YES})$ levels, which could distort the raw DiD through regression to the mean. The base-rate-normalized analysis in Figure 4 addresses this directly.

4.6 Base-Rate-Normalized Analysis

The base-rate normalized values are -0.0372 (95% CI $[-0.0812, +0.0068]$, nytimes.com, $n = 257$), -0.0476 (95% CI $[-0.0935, -0.0016]$, cnn.com, $n = 253$), -0.0394 (95% CI $[-0.1034, +0.0246]$, nypost.com, $n = 114$), and -0.0485 (95% CI $[-0.1026, +0.0055]$, foxnews.com, $n = 197$). All sites show negative δ_{norm}, implying that the differences in the prediction shift are somewhat aligned with the degradation of accuracy relative to the control. The effect is only statistically significant for cnn.com (CI: $[-0.0935, -0.0016]$); the remaining three sites have CIs spanning zero.

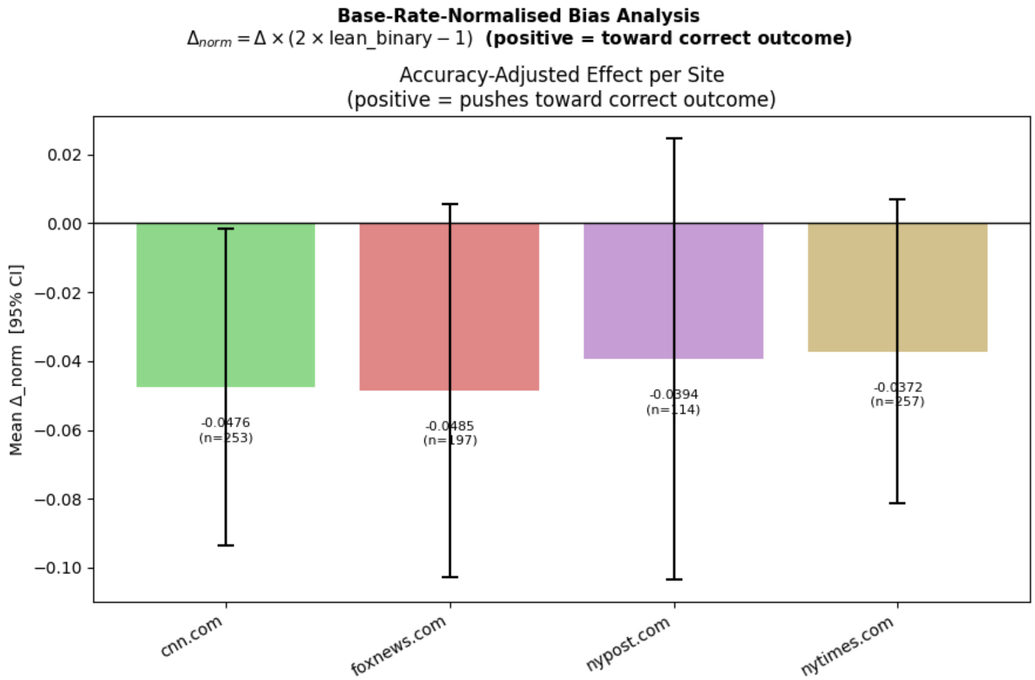


Figure 4: Base-rate-normalized analysis. δ_{norm} per site, where positive values indicate the site pushes predictions toward the correct partisan outcome on average.

4.7 Summary of Findings

Across all five analyses—relative bias, prediction accuracy, neutral event influence, difference-in-differences, and base-rate-normalized delta—we find no statistically significant evidence that the ingestion of a single article from nytimes.com, cnn.com, nypost.com, or foxnews.com produces stronger partisan directional bias in the LLM's political event predictions relative to the control condition, the ingestion of 8 articles from search engines. We find some evidence that the control condition performs better than the experimental conditions. The LLM forecasting system appears robust to the introduction of individual articles from outlets with known partisan leanings, at least under the experimental conditions studied here.

5 Discussion

Go through each section and review. At first glance, it looks like Figure 1, Panel 1 shows that there is clear

References

A Appendix / supplemental material

[APPENDIX PLACEHOLDER]